

Worksheet — 1.2 Software & Software Development

OCR A Level Computer Science (H446) — Component 01, Section 1.2 (1.2.1 Operating systems · 1.2.2 Applications generation · 1.2.3 Software development · 1.2.4 Types of programming language)

Name: ____ Date: ____

Time: ~30 min. Check against the answer key at the end.

1. Key-term match

Write the letter of the correct definition next to each term.

Term	Answer		Definition
1. Virtual memory	_____	A	Excessive paging between RAM and disk that slows the system almost to a halt.
2. Interrupt	_____	B	Software emulating a complete computer, or running platform-independent intermediate code.
3. Disk thrashing	_____	C	Using secondary storage as if it were RAM to run more/larger programs than RAM allows.
4. ISR	_____	D	Dividing memory into fixed-size blocks (pages mapped to frames).
5. Paging	_____	E	A signal (hardware or software) that a device/process needs the processor's attention.
6. Virtual machine	_____	F	The OS code (handler) run to deal with one particular interrupt.

2. The four stages of compilation (in order)

Fill in the blanks. Write the stages in the correct order.

Stage 1 — _____ **analysis**: source code is broken into ____; whitespace and comments are removed; a ____ table is built.

Stage 2 — ____ **analysis (also called** ____): tokens are checked against the language's ____; a ____ tree is built; _____ errors are reported here.

Stage 3 — **Code** ____: the parse tree and symbol table are used to produce ____ / machine code.

Stage 4 — **Code** ____: the code is refined to run ____ or use less ____, while producing the same result.

3. Translators — complete the comparison table

	Compiler	Interpreter
When does translation happen?	_____	_____
Output produced?	_____	_____
Program execution speed	_____	_____
Source code needed to run?	_____	_____
How are errors reported?	_____	_____
Best used for...	_____	_____

Q. In one sentence, what does an **assembler** translate, and at roughly what ratio?

4. Match the methodology to the scenario

Methodologies: **Waterfall** · **Agile** · **Extreme Programming (XP)** · **Spiral** · **RAD**

Write the best-fit methodology next to each scenario.

- a) A bank is building a large, expensive air-traffic adjacent system where the cost of failure is huge; management wants formal **risk analysis** at every iteration. → _____
- b) A small project for a client whose requirements are **fixed, clear and unlikely to change**, and who wants full documentation at each stage. → _____
- c) A start-up needs a working version **fast** and is happy to refine **prototypes** with users; requirements are vague. → _____
- d) Safety-critical control software where **code quality is paramount**, so the team uses **pair programming** and test-driven development with a customer on-site. → _____
- e) An app whose requirements are **expected to change a lot**; the client wants to see and feed back on **working increments** regularly. → _____

5. Addressing modes

A program executes the instruction **LDA 20**. Memory contents are:

Address	Contents
20	35
35	99
99	12

State the value that is loaded into the accumulator under each addressing mode.

- a) **Immediate** addressing: value loaded = _____

b) **Direct** addressing: value loaded = _____

c) **Indirect** addressing: value loaded = _____

d) In one line, explain what **indexed** addressing uses to find the operand, and what it is typically used for.

6. Scheduling & interrupts — short answer

a) Name the scheduling algorithm that gives each process a fixed **time slice (quantum)** in turn, returning unfinished jobs to the back of the queue. State whether it is pre-emptive.

b) **Shortest Job First** minimises average waiting time. Give **one** disadvantage.

c) Define **pre-emptive** scheduling.

d) Put the interrupt-handling steps in order (write 1–6):

Step	Order (1–6)
Load the address of the correct ISR	_____
Pop the registers off the stack and resume the original task	_____
CPU checks the interrupt register at the end of the F-D-E cycle	_____
Run the ISR	_____
Push the registers (e.g. PC, ACC) onto the stack	_____
Finish the current instruction (if the interrupt is higher priority)	_____

e) Why must the registers be saved on a **stack** (LIFO) rather than in a single store?

7. OOP — terms labelling/matching

Match each OOP term to its meaning.

Terms: **Class** • **Object** • **Method** • **Attribute** • **Inheritance** • **Encapsulation** • **Polymorphism**

Meaning	Term
A specific instance of a class.	_____
A template/blueprint defining attributes and methods.	_____
A variable belonging to a class/object (its data/state).	_____
A subroutine belonging to a class (its behaviour).	_____
Bundling data and methods together and hiding the data (private attributes accessed via methods).	_____
A subclass acquiring the attributes/methods of a superclass, and able to add or override them.	_____
The same method name behaving differently depending on the object.	_____

8. Challenge box

Challenge. A team is distributing finished commercial software to customers who must **not** see the source code, and the program must **run quickly** on the user's machine without needing anything else installed.

(i) Should they use a **compiler** or an **interpreter**? Justify with **two** reasons drawn from the table in Q3.

(ii) The OS uses **virtual memory** to let several large programs run at once. Explain why virtual memory does **not** actually increase the amount of RAM, and name the problem caused by **excessive** swapping.
